

WEEK 3:

**Credit Card Fraud Detection Using
Machine Learning**

**A Detailed Machine Learning Model
Development and Evaluation Plan**

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COURSE:

**Bachelor of Computer Applications
(BCA)**

INTERNSHIP:

Data Science Internship

TECHNOLOGY: **Python and Machine
Learning**

INTRODUCTION

Credit cards are widely used for online and offline financial transactions. As digital payments increase, fraudulent transactions have also become an important concern for financial institutions and customers.

Credit card fraud occurs when a transaction is made without the legitimate cardholder's authorization. Fraudulent transactions can cause financial losses and reduce customer confidence in digital payment systems.

Machine Learning can help detect suspicious transactions by learning patterns from historical transaction data. A trained model can analyze transaction characteristics and classify transactions as potentially legitimate or fraudulent.

This project proposes a conceptual **Credit Card Fraud Detection System using Machine Learning and Python.**

The system will analyze transaction-related features and predict whether a transaction is likely to be fraudulent.

The project focuses on designing the complete Machine Learning workflow rather than implementing the model with actual code.

PROBLEM DEFINITION

2.1 Hypothetical Problem

The proposed system will classify credit card transactions into two categories:

- Legitimate Transaction
- Fraudulent Transaction

The system will learn patterns from historical transaction records and use those patterns to identify suspicious new transactions.

2.2 Input Data

Possible transaction-related features may include:

- Transaction amount
- Transaction time
- Transaction frequency
- Merchant category
- Transaction location
- Device information
- Previous transaction behavior
- Transaction history indicators
- Encoded transaction features

2.3 Target Variable

The target variable will be:

Fraud Status

0 = Legitimate

1 = Fraudulent

Therefore, this project is a binary classification problem.

2.4 Problem Justification

Fraud detection is suitable for Machine Learning because fraudulent transactions may

have complex patterns that are difficult to identify using fixed rules alone.

Machine Learning can analyze large amounts of historical transaction data and identify patterns associated with fraudulent behavior.

PROJECT OBJECTIVES

The major objectives of this project are:

- To understand the credit card fraud detection problem.
- To define a suitable binary classification framework.
- To identify relevant transaction features.
- To develop a systematic data preprocessing strategy.
- To handle missing values, outliers, and inconsistent records.
- To address the problem of highly imbalanced classes.

- To perform feature engineering and feature selection.
- To compare multiple Machine Learning algorithms.
- To select an appropriate final model.
- To define a suitable model training strategy.
- To perform hyperparameter tuning.
- To use cross-validation for robust validation.
- To evaluate the model using appropriate classification metrics.
- To design a conceptual deployment architecture.
- To establish a model monitoring and maintenance strategy.

Expected Main Goal

The main goal is to develop a Machine Learning framework that can identify potentially fraudulent transactions while minimizing incorrect classifications of legitimate transactions.

PROJECT SCOPE

In scope

The following activities are included:

- Problem definition
- Data understanding
- Data preprocessing
- Data cleaning
- Missing-value handling
- Outlier analysis
- Feature engineering
- Feature selection
- Handling class imbalance
- Model selection
- Model training
- Hyperparameter tuning
- Cross-validation
- Performance evaluation
- Model comparison
- Conceptual deployment
- Model monitoring

- Model maintenance

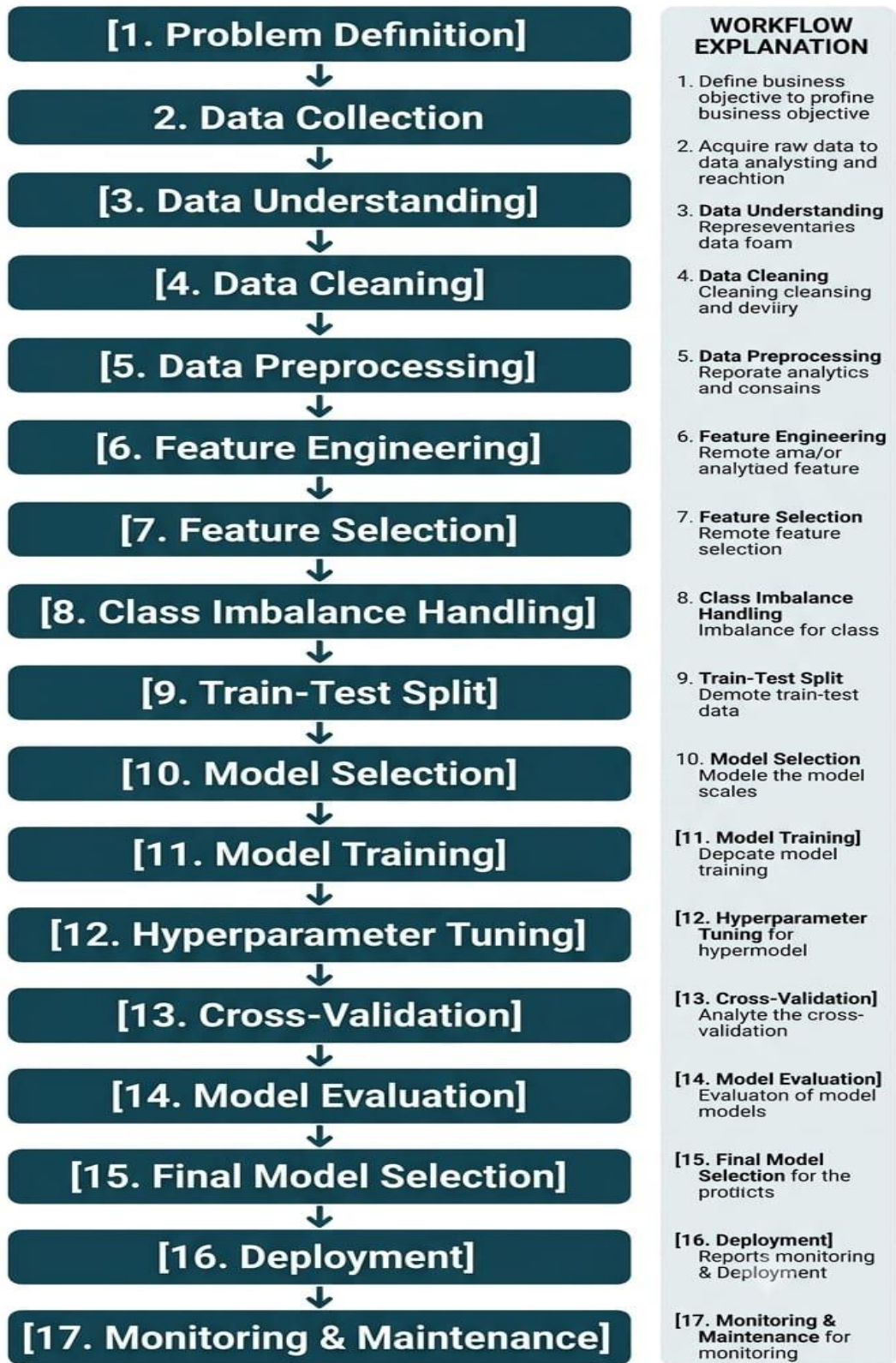
Out of Scope

The following activities are outside the scope of this project:

- Processing real customer financial information
- Accessing real banking systems
- Real-time financial transactions
- Actual financial decision-making
- Complete production deployment
- Direct connection to payment gateways
- Handling real customer identity information
- Performing actual financial investigations
- Guaranteeing that every fraudulent transaction will be detected

MACHINE LEARNING WORKFLOW

MACHINE LEARNING PROJECT WORKFLOW



Derived from <IMAGE 0>

Workflow Explanation

The workflow begins by defining the fraud detection problem. The available transaction data is then cleaned and transformed into a format suitable for Machine Learning.

Relevant features are selected, and the class imbalance problem is addressed. Multiple models are trained and evaluated using appropriate validation techniques.

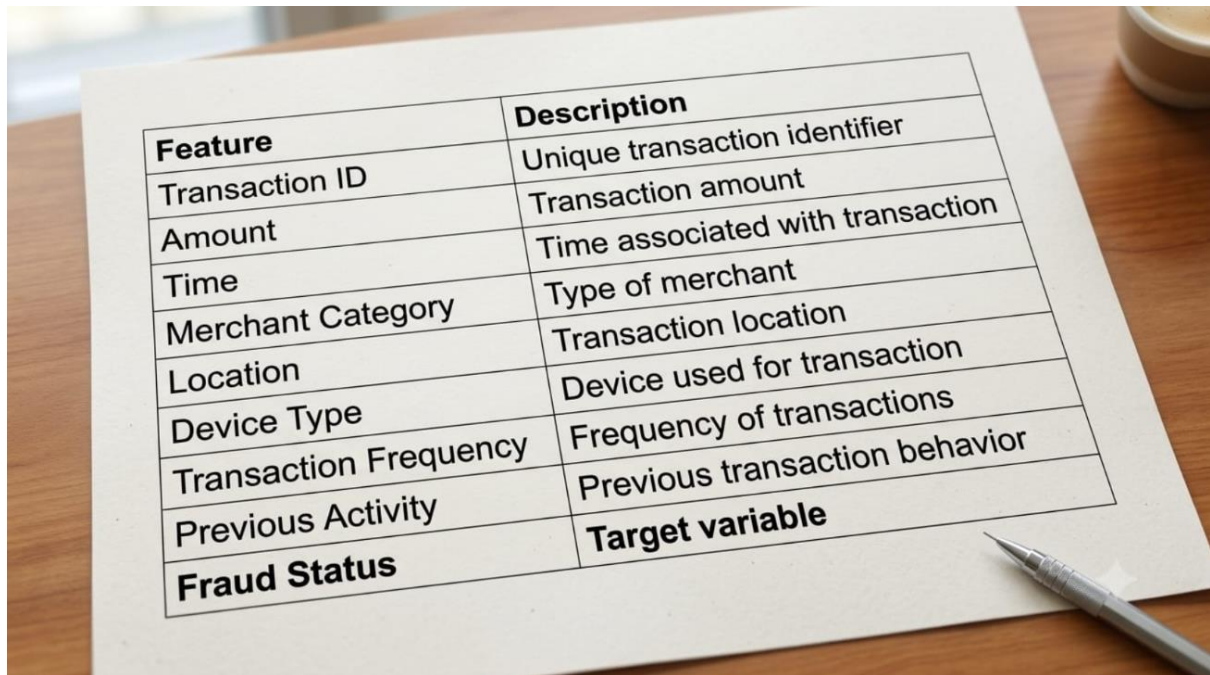
After selecting the best-performing model, it can be conceptually deployed and continuously monitored.

DATA UNDERSTANDING

Data understanding is an important step before preprocessing and model training.

The proposed dataset may contain thousands or millions of transaction records.

Possible Data Attribute_j



Feature	Description
Transaction ID	Unique transaction identifier
Amount	Transaction amount
Time	Time associated with transaction
Merchant Category	Type of merchant
Location	Transaction location
Device Type	Device used for transaction
Transaction Frequency	Frequency of transactions
Previous Activity	Previous transaction behavior
Fraud Status	Target variable

Important Observation

Fraud transactions are usually much fewer than legitimate transactions. Therefore, the dataset is expected to be highly imbalanced.

For example, a hypothetical dataset might contain:

99% legitimate transactions Fraudulent

1% fraudulent transactions

DATA PREPROCESSING

Data preprocessing converts raw transaction data into a clean and usable form.

Data Cleaning

The following activities will be performed:

- Identify missing values.
- Remove duplicate records.
- Check incorrect data types.
- Identify invalid transaction amounts.
- Check inconsistent categories.
- Verify feature ranges.
- Detect unusual observations.

Missing Values

Missing information can reduce model reliability.

Possible solutions include:

- Mean or median imputation for numerical features.
- Mode imputation for categorical variables.
- Appropriate missing-value indicators.
- Removing records only when justified.

Duplicate Records

Duplicate transaction records can cause the model to learn the same observation multiple times.

Therefore, duplicate records should be identified and handled before model training.

Outlier Handling

Outliers should not automatically be removed because unusual transaction

values may themselves be indicators of fraud.

Outliers should first be investigated using:

- IQR method
- Z-score
- Distribution analysis
- Domain knowledge

FEATURE SELECTION AND SCALING

Feature Selection

Not every available feature will necessarily improve the model.

Feature selection can be performed using:

- Correlation analysis

- Statistical methods
- Feature importance
- Recursive Feature Elimination
- Domain knowledge
- Removing irrelevant features can:
 - Reduce model complexity
 - Improve training efficiency
 - Reduce noise
 - Improve generalization

Feature Scaling

Some Machine Learning algorithms are sensitive to differences in feature scales.

Possible techniques include:

- Standardization
- Min-Max Scaling
- Robust Scaling

Scaling should be performed carefully using only training data information to prevent data leakage.

CLASS IMBALANCE

Handling Class Imbalance

Class imbalance is one of the most important challenges in credit card fraud detection.

If legitimate transactions greatly outnumber fraudulent transactions, a model could achieve high accuracy simply by predicting most transactions as legitimate.

For example:

If 99 out of 100 transactions are legitimate, a model that predicts every transaction as legitimate could achieve 99% accuracy but would detect **zero fraud cases**.

Therefore, accuracy alone is not sufficient.

Possible Solutions

1. Class Weighting

The model can assign greater importance to the minority fraud class.

2. Oversampling

The minority class can be increased using techniques such as SMOTE.

3. Undersampling

Some majority-class observations can be reduced.

4. Appropriate Evaluation Metrics

Greater importance should be given to:

- Precision
- Recall

The selected strategy should be tested carefully to avoid introducing additional bias or data leakage

MODEL SELECTION

Several classification algorithms will be considered.

Logistic Regression

Logistic Regression provides a simple baseline for binary classification.

Advantages:

- Easy to understand.
- Fast to train.
- Provides a strong baseline.

Limitation:

May not capture highly complex relationships.

Decision Tree

Decision Tree models classify observations using a series of decision rules.

Advantages:

- Easy to interpret.
- Can model nonlinear relationships.
- Handles different types of features.
- Limitation:
- Can overfit without proper control.

Random Forest

Random Forest combines multiple decision trees.

Advantages:

- Robust for structured data.
- Handles nonlinear relationships.
- Provides feature importance.

- Usually more stable than a single decision tree.

Limitation:

- Less interpretable.
- Can require more computational resources.

Gradient Boosting

Gradient Boosting builds models sequentially to improve prediction errors.

Advantages:

- Strong predictive performance.
- Can capture complex relationships.

Limitation:

- Requires careful tuning.
- Can be computationally expensive.

CHOSEN MODEL

Proposed Primary Model: Random Forest

Random Forest will be considered as the primary model for the proposed fraud detection system.

Reasons for Selection

1. It can capture nonlinear relationships.
2. It works well with structured/tabular data.
3. It can handle many input features.
4. It provides feature importance information.
5. It is generally robust against noise.
6. It can reduce the overfitting risk associated with individual decision trees.

However, the final model should be selected based on actual validation results rather than assuming Random Forest will always be the best model.

Proposed Model Comparison

Model	 Strength	 Limitation
 Logistic Regression	Simple and interpretable  	Limited complex pattern detection
 Decision Tree	Easy to understand	Can overfit 
 Random Forest	Robust and flexible	Less interpretable
 Gradient Boosting	Strong predictive performance  	Requires tuning 

MODEL TRAINING PROCESS

The model training process will follow these steps:

Step 1 – Prepare Data

Use the cleaned and transformed dataset.

Step 2 – Separate Features and Target

Input features will be separated from the fraud-status target variable.

Step 3 – Split the Dataset

A training and testing set will be created.

A possible initial split is:

- 80% Training
- 20% Testing

Stratified splitting should be used to maintain the fraud-to-legitimate ratio.

Step 4 – Train Baseline Model

Logistic Regression can be used as a baseline.

Step 5 – Train Candidate Models

Decision Tree, Random Forest, and Gradient Boosting models can be trained.

Step 6 – Perform Cross-Validation

Cross-validation will be used to estimate generalization performance.

Step 7 – Tune Hyperparameters

The selected model will be optimized.

Step 8 – Final Testing

The final model will be evaluated on the untouched test dataset.

HYPERPARAMETER TUNING

Hyperparameters control the behavior of a Machine Learning algorithm and must be selected before training.

For Random Forest, possible hyperparameters include:

- Number of estimators
- Maximum tree depth
- Minimum samples required for splitting
- Minimum samples required at a leaf
- Maximum number of features
- Class-weight settings

Grid Search

Grid Search tests a predefined set of parameter combinations.

Advantage: Systematic search.

Limitation: Can become computationally expensive.

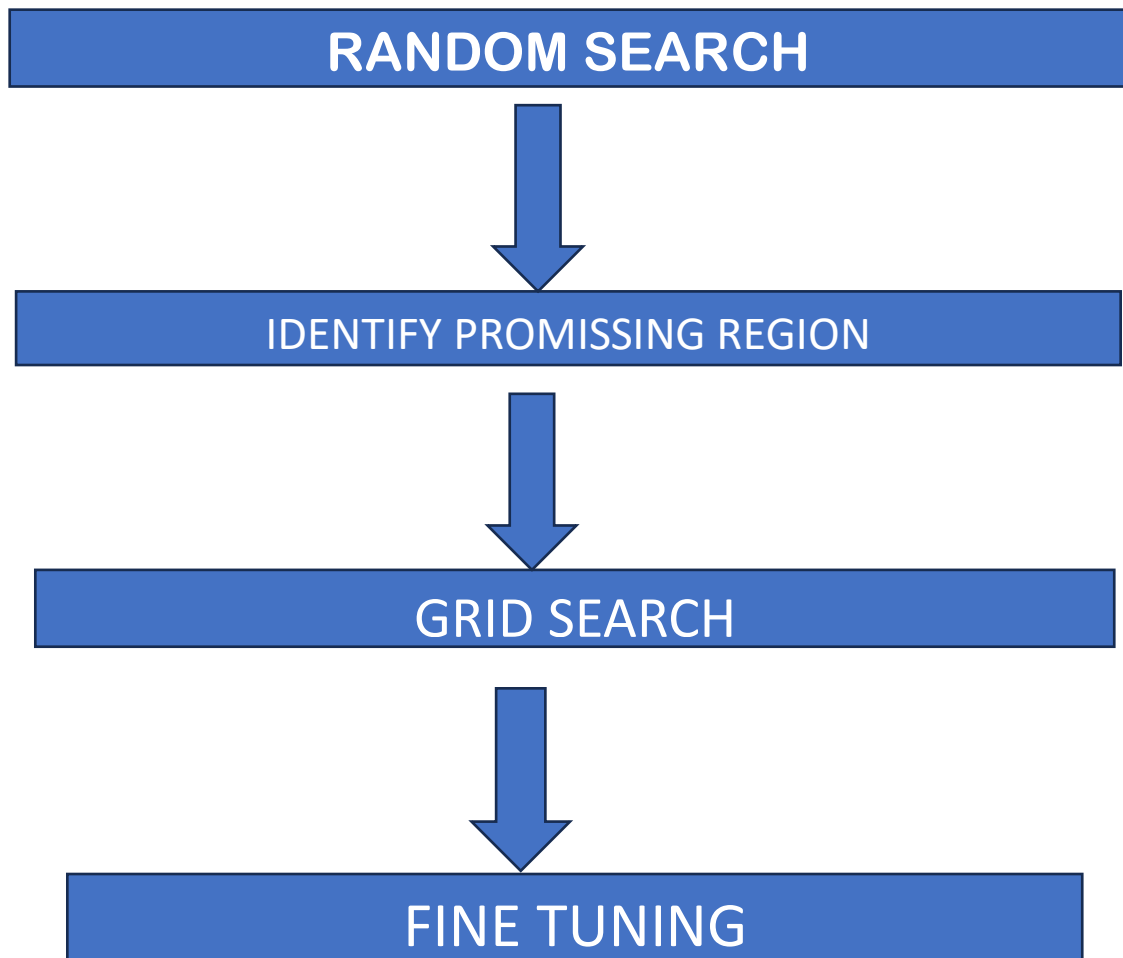
Random Search

Random Search evaluates randomly selected parameter combinations.

Advantage: Can explore a larger parameter space more efficiently.

Proposed Strategy

A two-stage approach can be used:



This approach can provide
an efficient balance
between exploration and
detailed optimization.

MODEL EVALUATION METRICS

Fraud detection requires careful metric selection because fraudulent transactions are usually the minority class.

Accuracy

Accuracy represents the percentage of correctly classified transactions.

However, accuracy alone can be misleading when the dataset is highly imbalanced.

Precision

Precision measures the proportion of transactions predicted as fraud that are actually fraudulent.

High precision helps reduce false fraud alerts.

Recall

Recall measures the proportion of actual fraudulent transactions that are successfully detected.

Recall is particularly important in fraud detection because missed fraud cases can cause financial loss.

F1-Score

F1-score provides a balance between precision and recall.

It is useful when both false positives and false negatives matter.

ROC-AUC

ROC-AUC measures the model's ability to distinguish between legitimate and fraudulent transactions across classification thresholds.

Precision-Recall AUC

Because fraud datasets can be highly imbalanced, Precision-Recall AUC can provide additional insight into minority-class performance.

Confusion Matrix

The confusion matrix contains:

- True Positive
- True Negative
- False Positive
- False Negative

Together, these metrics provide a more complete evaluation of fraud detection performance.

VALIDATION STRATEGY

Cross-Validation and Validation

Validation is essential for determining whether the model performs consistently on unseen data.

Train-Test Split

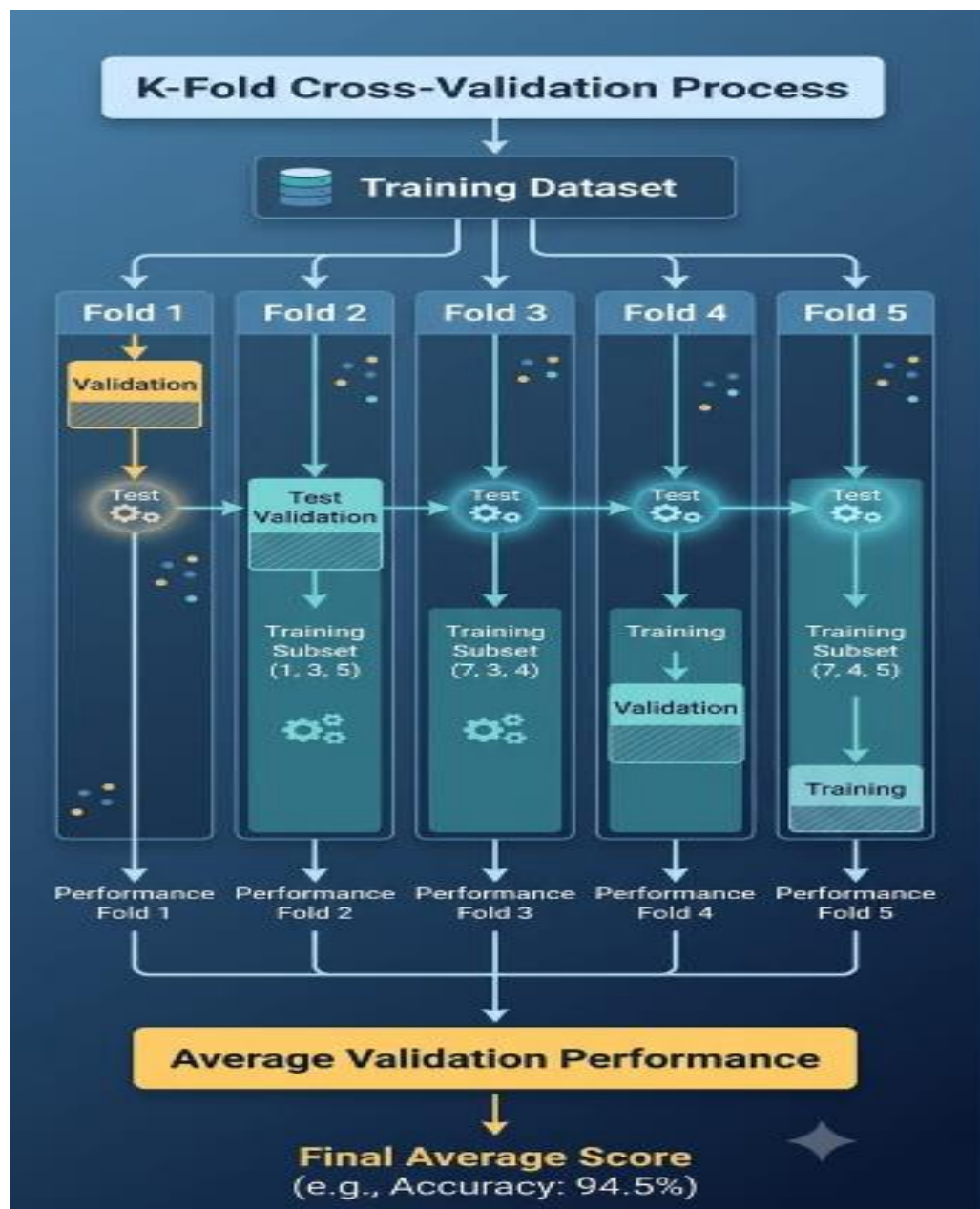
The dataset will initially be divided into training and testing portions.

The test set should remain untouched until final evaluation.

Stratified K-Fold Cross-Validation

For classification, Stratified K-Fold Cross-Validation can be used.

The method maintains approximately similar class proportions across folds.



Data Leakage Prevention

Preprocessing steps should be fitted only on the relevant training data within each validation fold.

Oversampling techniques such as SMOTE should also be applied only to training folds, never before splitting the data.

This prevents information from the validation/test data from influencing the model

MODEL EVALUATION AND COMPARISON

Model Evaluation Process

After training and tuning, the candidate models will be compared.

Evaluation Procedure

- Establish Logistic Regression as a baseline.
- Train Decision Tree.
- Train Random Forest.
- Train Gradient Boosting.
- Perform stratified cross-validation.
- Calculate precision, recall, F1-score, ROC-AUC and PR-AUC.

- Analyze confusion matrices.
- Compare validation performance.
- Select the most appropriate model.
- Evaluate the final model on the untouched test set.

Model Selection Priority

For fraud detection, the model should not be selected only by accuracy.

The selection should consider:

Recall + Precision + F1-Score + ROC-AUC + PR-AUC + Business Cost

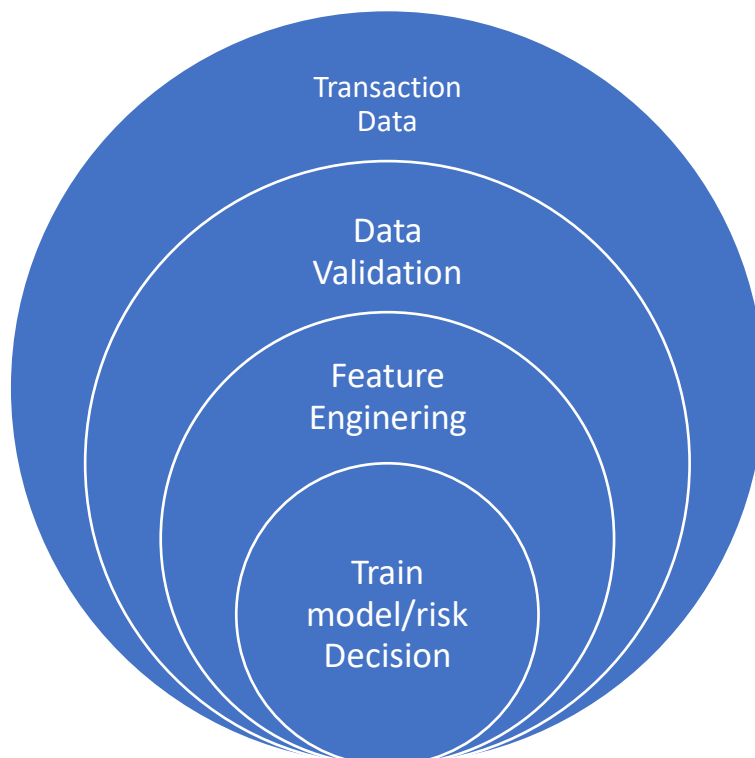
A very high recall model may detect more fraud but could also generate many false alerts. Therefore, an appropriate balance must be established.

DEPLOYMENT PLAN

Conceptual Model Deployment

Actual deployment is optional for this assignment, but a future deployment architecture can be proposed.

Deployment workflow



Possible Technologies

- Python
- Pandas
- NumPy
- Scikit-learn
- Flask or FastAPI
- Database system
- Cloud/server infrastructure
- Monitoring tools

The trained model could generate a fraud probability for each transaction. Transactions with high-risk scores could be sent for additional verification or manual review.

MODEL MONITORING AND MAINTENANCE

Model Monitoring and Maintenance

A Machine Learning model may become less effective over time because transaction patterns and fraud techniques can change.

Performance Monitoring

Monitor:

Precision

Recall

F1-score

ROC-AUC

PR-AUC

False-positive rate

False-negative rate

Data Drift

Data drift occurs when the characteristics of new transaction data differ from the data used during model training.

The system should monitor important feature distributions.

Concept Drift

Fraud patterns can change over time. A pattern that was previously associated with fraud may become less useful later.

Model Retraining

The model should periodically be retrained using recent, verified transaction data.

Model Versioning

Different model versions should be tracked so that the system can return to a previous stable model if necessary.

Security and Privacy

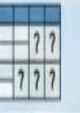
Financial data is sensitive. Any real implementation should use:

- Data encryption
- Access controls
- Secure storage
- Data minimization
- Anonymization where appropriate

CHALLENGES AND SOLUTIONS

Challenges and Proposed Solutions

CHALLENGES AND PROPOSED SOLUTIONS IN FRAUD DETECTION

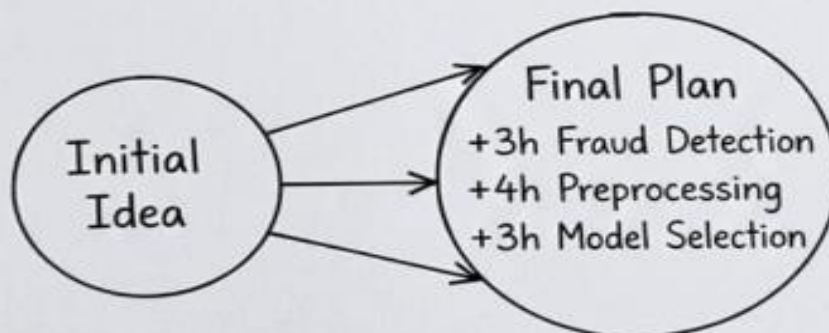
CHALLENGE		PROPOSED SOLUTION
 HIGHLY IMBALANCED DATASET 		• CLASS WEIGHTING & RESAMPLING: Give minority classes more weight. Carefully resample data. • APPROPRIATE METRICS: Use F1-score, Precision-Recall AUC (PR-AUC) instead of accuracy.
 MISSING VALUES 		• IMPUTATION OR HANDLING: Fill missing values with mean, median, mode, or use algorithms that handle them natively.
 DUPLICATE TRANSACTIONS 		• DUPLICATE DETECTION & REMOVAL: Identify and remove identical transactions to prevent bias and wasted computation.
 OUTLIERS 		• INVESTIGATE RATHER THAN REMOVE: Determine if outliers are errors or valuable indicators of fraud before removing.
 OVERFITTING 		• CROSS-VALIDATION & TUNING: Use techniques like K-fold cross-validation and optimize hyperparameter settings.
 DATA LEAKAGE 		• PROPER PREPROCESSING PIPELINES: Ensure data transformation happens within each fold of cross-validation.
 FALSE POSITIVES 		• OPTIMIZE PRECISION & THRESHOLD: Balance recall by increasing precision and adjusting classification thresholds.
 MISSED FRAUD 		• FOCUS ON RECALL & THRESHOLD: Minimize false negatives by lowering thresholds to capture more fraud cases.
 CHANGING FRAUD PATTERNS 		• CONTINUOUS MONITORING & RETRAINING: Monitor model performance regularly and retrain with fresh data. Adapt features.
 PRIVACY RISKS 		• ENCRYPTION & ACCESS CONTROL: Implement strong data encryption and strict access permissions to protect sensitive information.
 LARGE TRANSACTION VOLUME 		• EFFICIENT PROCESSING INFRASTRUCTURE: Utilize big data technologies (e.g., Spark, cloud databases) and efficient algorithms.
 MODEL INTERPRETABILITY 		• FEATURE IMPORTANCE & EXPLAINABILITY METHODS: Generate insights using feature importance scores and explainable AI techniques.

30–35 HOUR WORK PLAN

Proposed Project Time Allocation

The assignment recommends approximately 30–35 hours of work.

Activity	Estimated Time (hours)
Problem definition and research	3h
Understanding fraud detection	3h
Data preprocessing planning	4h
Feature engineering and selection	3h
Class imbalance analysis	3h
Model selection	4h
Training strategy	3h
Evaluation metrics	3h
Validation strategy	3h
Deployment and maintenance	2h
Diagram preparation	2h
Documentation and final review	3h
Total	(36h)



MACHINE LEARNING WORKFLOW DIAGRAM

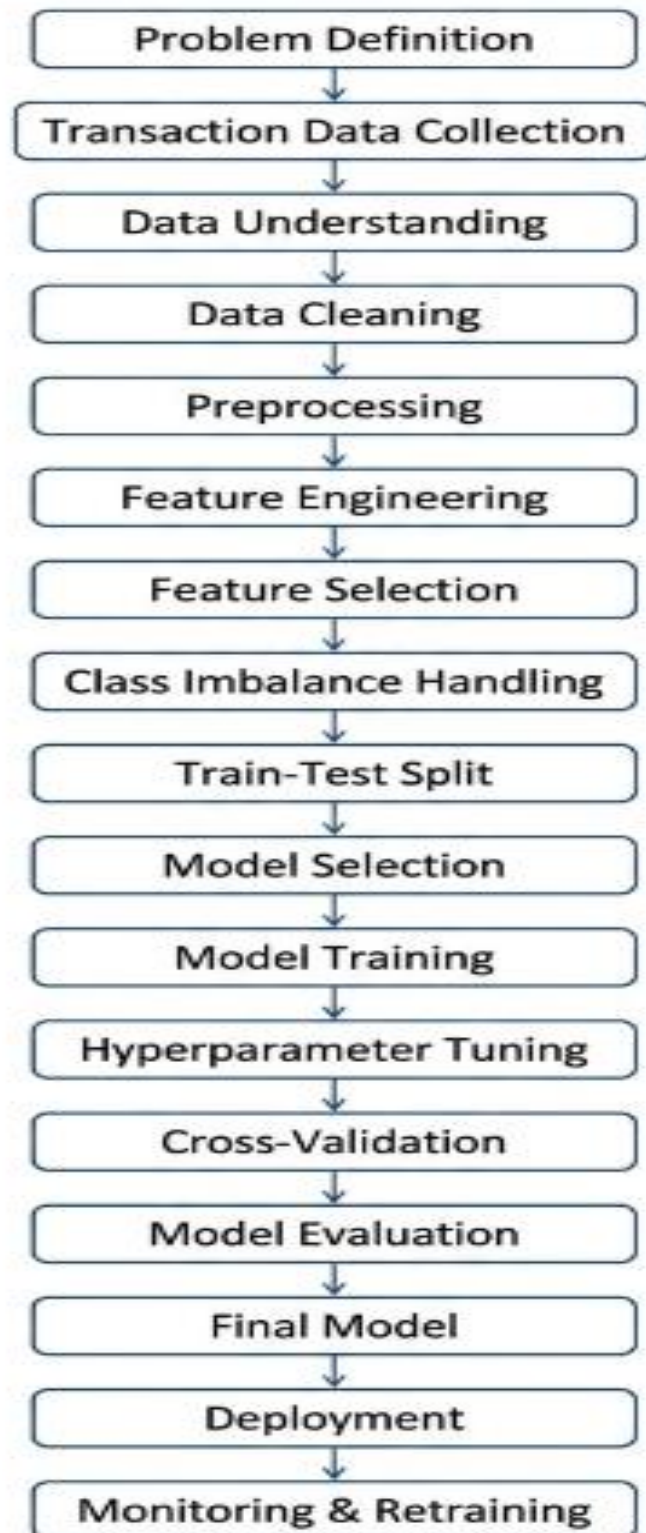
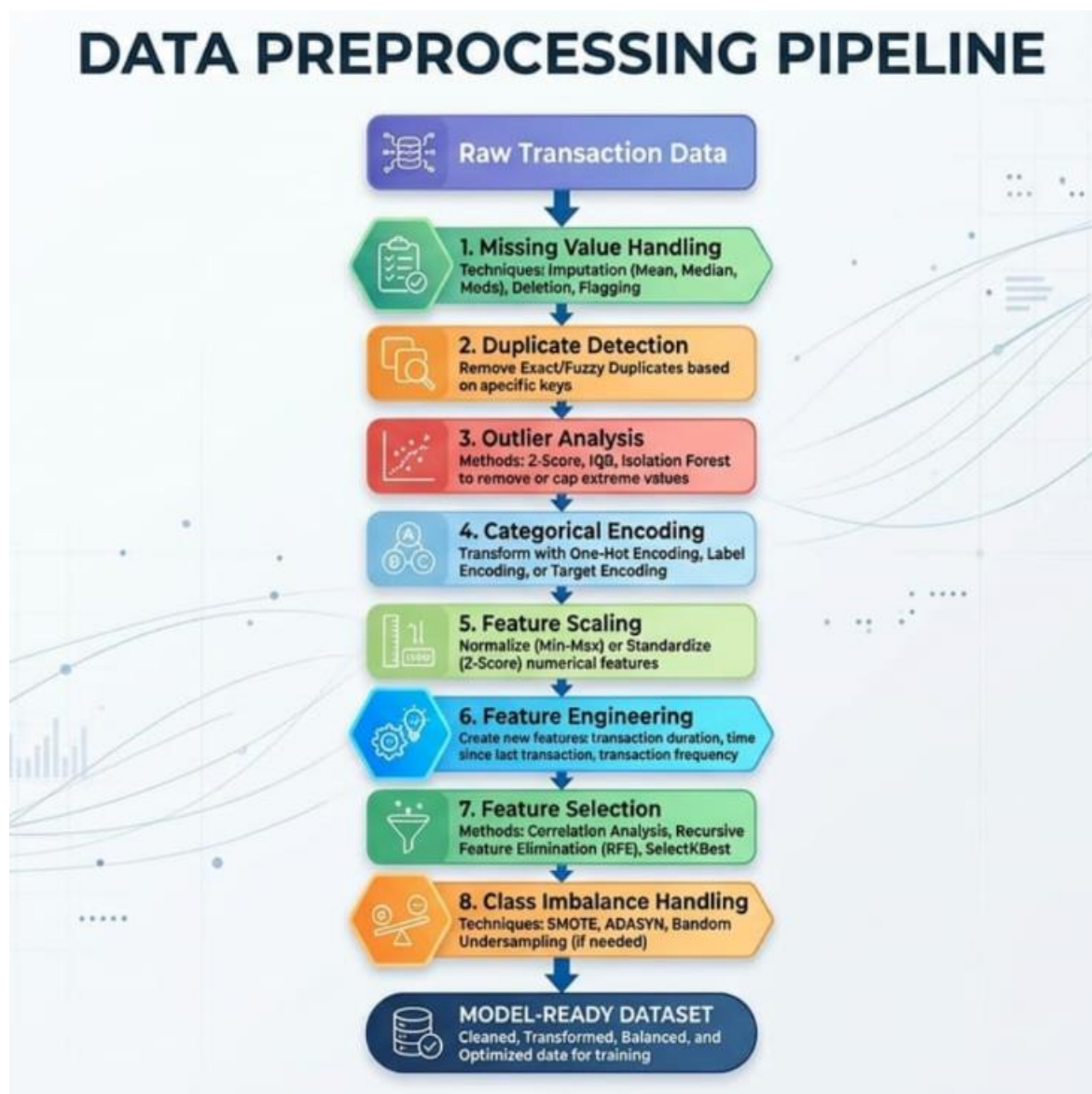


Diagram Explanation

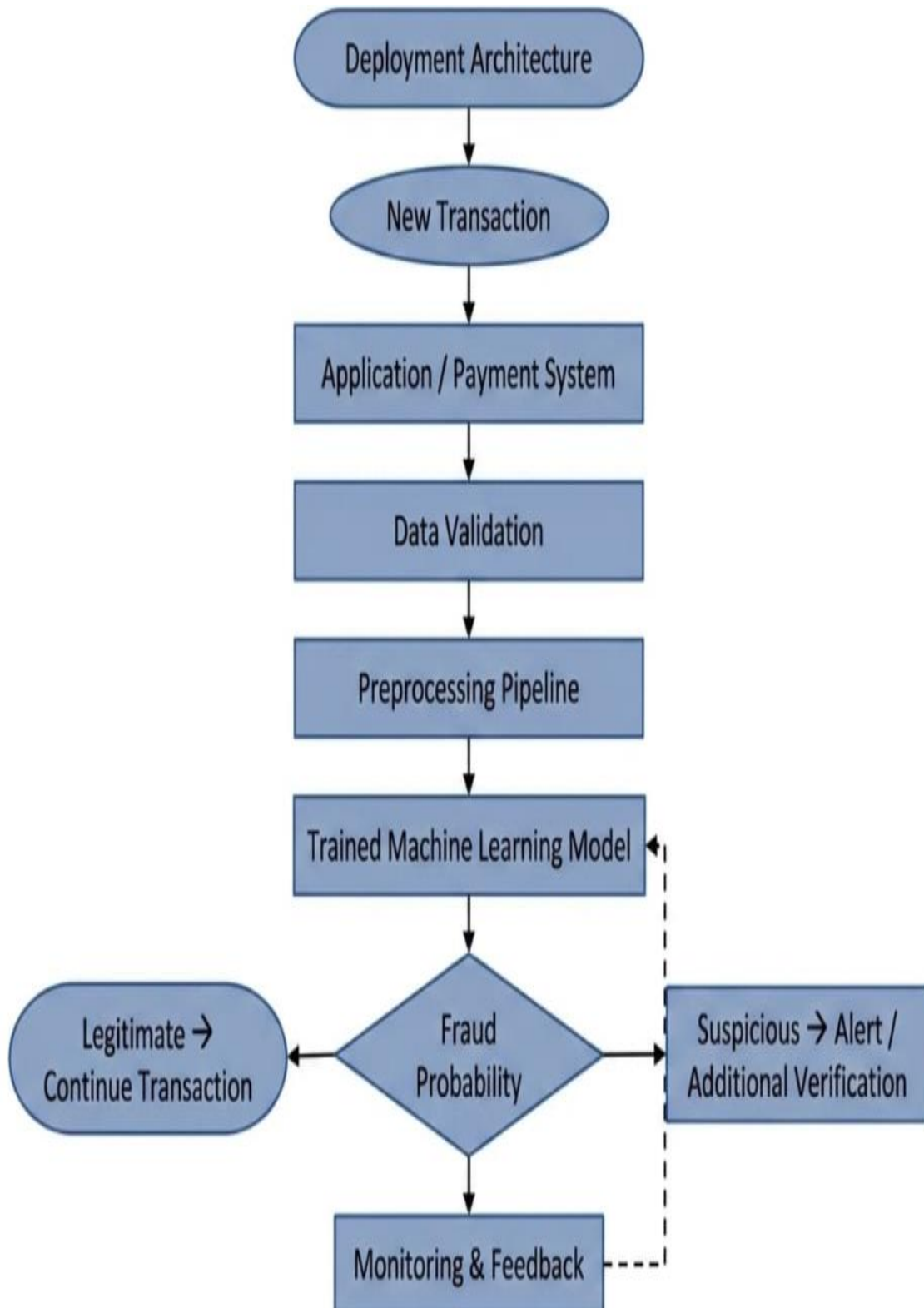
This diagram represents the complete lifecycle of the proposed Machine Learning fraud detection system.

PREPROCESSING AND DEPLOYMENT DIAGRAM



Deployment Architecture

DIGNAM 3



CONCLUSION

This Week 3 project provides a comprehensive conceptual plan for developing a **Credit Card Fraud Detection System using Machine Learning and Python.**

The project begins with a **clearly defined binary classification problem** and covers all major stages of **Machine Learning development.**

The proposed workflow includes **data cleaning, preprocessing, feature engineering, feature selection, class imbalance handling, model selection, model training, hyperparameter tuning, cross-**

validation, and performance evaluation.

Several algorithms, including **Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting**, can be compared. Random Forest is proposed as the primary candidate because it can capture complex relationships and works effectively with structured data. However, the final selection should be based on **validation and test performance**.

Because fraudulent transactions are usually a minority class, accuracy alone should not be considered sufficient. **Precision, Recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix analysis** should be

used to understand the model's performance.

The proposed deployment and maintenance plan also demonstrates how the model could be monitored and retrained when transaction patterns change.

Overall, this project provides a realistic strategic framework that can later be implemented as a complete **Python-based Machine Learning application**.

